23MAT112
Mathematics for Intelligent Systems – 2
Practice Sheet – 6
Sampling from Univariate Distributions

Sampling from Uniform Distribution

- rand(n,1)
  Generates n uniformly distributed numbers between 0 and 1
- a + (b-a)*rand(n,1)
  Generates n uniformly distributed numbers between a and b
- x=rand(n,1);histogram(x,m)
  Plots a histogram of n uniformly distributed random variables with m bins

1. Generate 100 uniformly distributed random numbers between 0 and 1 and draw a histogram with 10 bins
2. Generate 1000 uniformly distributed random numbers between 5 and 10 and 1 and draw a histogram with 10 bins.
3. Generate 500 uniformly distributed random numbers between -3 and 3.

(a) Find the mean and variance of the sample and verify if it is close to the mean and variance of the population.
(b) Draw the histogram of the sample

Sampling from Exponential Distribution

A sample from an exponentially distributed population can be obtained from a uniformly distributed sample in (0,1) using:

$$x = -\left(\frac{1}{\lambda}\right) \times ln(rand())$$

- Generation of a sample of 10000 exponentially distributed random numbers with mean 0.5. (lambda=2)

```
>>y=rand(10000,1);
>>lambda=2;
>>x=(-1/lambda)*log(y);
>>histogram(x,100)
```

1. Generate 1000 numbers from an exponentially distributed population with λ=1 and draw a histogram with 50 bins. Also find the mean and variance of the sample and compare it with the mean and variance of the population.
2. Generate 500 numbers from an exponentially distributed population with λ=3.
   (a) Draw a histogram with 20 bins and the pdf of the distribution in the two different figures and compare.
   (b) Find the mean and variance of the sample and the population and compare.

Sampling from Standard Normal/Gaussian N(0,1) Distribution using Box-Muller's formula

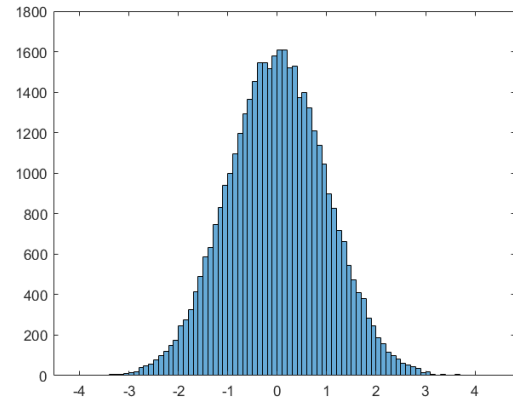
Let U_1 and U_2 be two independent uniformly distributed random number. Two Gaussian N(0,1) random numbers are

$$y_1 = \sqrt{-2\,ln(U_1)}\,Cos(2\pi U_2) \;\; and$$

$$y_2 = \sqrt{-2\,ln(U_1)}\,Sin(2\pi U_2)$$

- % generate 40000 values from N(0,1)

```
u1=rand(20000,1);
u2=rand(20000,1);
r= -2*log(u1);
theta=2*pi*u2;
x1=(sqrt(r)).*cos(theta);
x2=(sqrt(r)).*sin(theta);
x=[x1 x2];
histogram(x);
```



Sampling from Normal/Gaussian N($\mu = 5, \sigma = 2$) Distribution using Box-Muller's formula

```
% generate 40000 values from N(5,2)
clf;clc;
mu=5;
sigma=2;
u1=rand(20000,1);
u2=rand(20000,1);
r= -2*log(u1);
theta=2*pi*u2;
x1=mu+sigma*(sqrt(r)).*cos(theta);
x2= mu+sigma*(sqrt(r)).*sin(theta);
x=[x1 x2];
histogram(x);
```

Sampling from Normal/Gaussian N($0, 1$) Distribution using randn

```
>>x=randn(40000,1);
>>histogram(x)
```

Sampling from Normal/Gaussian N($\mu = 5, \sigma = 2$) Distribution using randn

```
mu=5
sigma=2
x=mu+sigma*randn(40000,1);
histogram(x)
```

OR

```
mu=5;
sigma=2;
size=[40000,1];
x=normrnd(mu,sigma,size);
histogram(x)
```

Generation of random numbers for discrete random variables

```
% Generation of random sample of size 1000 from distribution with the pmf:
%f(1)=0.5, f(2)=0.3, f(3)=0.2
n=1000 %size of the sample
x=rand(n,1)
  for i=1:n
     if x(i)<= 0.5
       x(i)=1;
     elseif x(i)<=0.8
            x(i)=2;
     else x(i)=3;
end
end
x
histogram(x)
```

To generate a sample of size 10000 from a binomial population with n=5 and p=0.5.

```
y=0:1:5;
f=pdf('bino',y,5,0.5);
F=cdf('bino',y,5,0.5);
n=10000 %sample size
x=rand(n,1)
    for i=1:n
        if x(i)<= F(1)
          x(i)=y(1);
        elseif x(i)<=F(2)
               x(i)=y(2);
        elseif x(i)<=F(3)
                x(i)=y(3);
        elseif x(i)<=F(4)
               x(i)=y(4);
        elseif x(i)<=F(5)
               x(i)=y(5);
        else x(i)=y(6);
       end
    end
x, histogram(x,'normalization','probability')
```

Evaluation of value of π computationally with the help of random number generation.

Concept: Area of [0 1]x[0 1] box =1;

Area of part of unit circle $x^2 + y^2 = 1$ in the first quadrant $= \frac{\pi}{4}$.

If 2D points (x,y) are randomly drawn from the [0,1]x[0,1] box,

probability that it is inside the circle $= \frac{\pi}{4}/1 = \frac{\pi}{4}$

```
% Evaluation of pi
N=100000;
x=rand(N,1);
y=rand(N,1);
rsq=x.^2+y.^2;
z=(rsq<=1);
pivalue=4*sum(z)/N
```

Practise Questions

1. Generate a sample of 2000 numbers from a standard normal population using Box-Muller's method and draw the histogram.
2. Generate a sample of 100 normally distributed numbers with mean 55 and standard deviation 10 using Box-Muller's formula and draw the histogram.
3. Generate 1200 numbers from a standard normal population, directly using the MATLAB command, 'randn'. (a) Draw a histogram and check if the curve is bell shape. (b) Find the mean and variance of the sample drawn and compare it with the mean and variance of the population.
4. Generate 500 numbers from a normal population with mean 300 and variance 50, using 'randn' command.

(a) Draw a histogram with 20 bins.
(b) Find the mean and variance of the sample and compare with that of the population .
(c) Plot the pdf of the normal distribution.

5. Evaluate the mean and variance of the samples generated in the previous 2 questions and compare it with the actual mean and variance of the given distribution.
6. Generate a sample of size 1000 from a population with probability density, $f(x)=3x^2$, $0<x<1$ and draw the histogram for the sample with 20 bins. Also find the sample mean, sample variance of the generated sample and compare with the actual mean and variance of the population. (Hint: Use CDF in the way it was used to find the formula to generate sample for exponential distribution)
7. Generate a sample of size 500 from a population with probability mass function, $f(5)=0.2$, $f(9)=0.1$, $f(12)=0.4$, $f(15)=0.3$.
8. Generate a sample of size 100 from a binomial population with $n=3$ and $p=0.2$. Show the histogram with probability.
9. Generate a sample of size 1000 from a binomial population with $n=12$ and $p=0.25$. Show the histogram with probability. Evaluate the mean and variance of this sample and compare with the mean and variance of the actual population.
10. Try to come up with another method to find the value of 'pi' using random number generation and probability.